UNIVERSITY OF CAMBRIDGE

DiS MPHIL

Supplementary Material

Mechanistic Interpretability of Open Source LLMs

Reproduction of On the Biology of a Large Language Model

IULIIA VITIUGOVA

MPHIL IN DATA INTENSIVE SCIENCE, UNIVERSITY OF CAMBRIDGE

These appendices accompany the main report. References to sections, tables, and equations point to it, and every figure and table here is reproducible from the committed result files in the repository.

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S.A Orthogonality in high dimension

A small cosine between two directions is not on its own evidence of orthogonality, because in high dimension almost any two directions are close to orthogonal. For isotropically distributed unit directions in $\mathbb{R}^d$ the expected absolute cosine is

$$\mathbb{E}|\cos| \approx \sqrt{\tfrac{2}{\pi d}}, \tag{1}$$

which for $d = 2560$ is about 0.016. A small cosine is therefore expected under isotropy and is not by itself evidence of a structured relationship. Real representations are anisotropic, so the isotropic baseline is not the appropriate null, and we compare the measured cosine not against zero or the isotropic baseline but against a null distribution of cosines matched to the covariance of the activations, which accounts for the anisotropy of the representation [1, 2].

The effective dimension of the representation is estimated from the participation ratio of the eigenvalues of the activation covariance,

$$d_{\text{eff}} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}. \tag{2}$$

Its value lies well below the nominal $d = 2560$, of order 13 in the decision layers and between about 3 and 20 across all layers. Against this covariance-matched null the reading and use directions are orthogonal. The measured $\cos(w_{\text{LDA}}, u)$ lies at about the 18th percentile of the null, below its median, and does not exceed the 95th percentile on any of the 36 layers. The measured cosine would exceed the isotropic threshold, but this reflects the inadequacy of the isotropic null under anisotropy rather than an alignment of the two directions.

Because the reading direction is estimated with finite reliability ($r_w \approx 0.72$ to 0.78, against $r_u \approx 0.99$ for the use direction), we apply the standard correction for attenuation, which gives an upper bound on the true correlation,

$$\rho_{\text{true}} \leq \frac{\rho_{\text{obs}}}{\sqrt{r_w r_u}}. \tag{3}$$

The per-layer upper bound on the correlation between reading and use never exceeds 0.074, and at the typical cosine it is a few hundredths. This bound is the main quantitative statement of the orthogonality, and the primary evidence that reading and use are distinct directions is the causal dissociation of Section 3.1.2, for which the geometric orthogonality is corroboration. For every reported cosine the measured value, the null median, the significance threshold (95th percentile), the null type, and the effective dimension are stated, so that each orthogonality is evaluated against the anisotropy of the representation rather than the isotropic baseline.

S.B Decision zone

The reading, writing, and use directions all change most sharply within a narrow band of layers rather than uniformly across depth. This appendix collects the evidence that the model commits to its answer within a specific band, which we call the decision zone, and locates its centre.

We take a set of measurements from different families, geometric, dictionary, feature, attribution, and causal. Each of them rises gradually with depth, and for each we record the layer at which it first exceeds its null. The null of a measurement is the value it takes under a matched random control, a random direction or a random set of features of the same size. Table 1 lists the measurements together with what each one shows.

measurement	family	onset	value
the two classes pull apart along the use direction by more than 1σ	geometric	L19	1.72σ
a linear classifier on the residual reaches AUC 0.96	geometric	L21	AUC 0.96
the writing direction enters the span of the active dictionary features	dictionary	L21	0.29 capture
the use direction enters the span of the active dictionary features	dictionary	L21	0.12 capture
dictionary features aligned with the use direction rise above the null	feature	L21	11 to 30
the attribution over the active features exceeds a matched random set	attribution	L21	$1.10\times$ null
steering along the use direction makes the target class token the top prediction	causal	L21–24	full

Table 1: Layer at which each measurement first exceeds its null, on the α/β decay corpus.

The measurements rise through the band from L19 to L24, and most of them reach their level at L21. The two classes begin to pull apart along the use direction (Section 2.5.3) at L19. By L21 the remaining measurements have risen above their nulls. A linear classifier on the residual reaches an AUC of 0.96, the full decodability of Section 3.1, whereas the projection onto the use direction alone remains close to chance (Table 1). The classes are therefore decodable from the residual without that decodability lying along the use direction. The writing and use directions come to lie in the span of the active dictionary features (the capture of Section 3.5), the number of dictionary features that write along the use direction grows, and the attribution over the active features rises above its random baseline (Section 2.8). The causal effect grows in step. Steering along the use direction (Section 2.7) has little effect in the early layers, with an intact rate of 0.06 at L18, rises through L19 and L20, and produces a full reversal from L21 to L22.

The same band appears on a second surface form. On the scaffold form the class means separate along the use direction at L19 and a linear classifier on the residual reaches an AUC of 0.9 at L21, matching the α/β decay corpus, which indicates that the decision zone is not specific to the surface form.

These layers agree with the main results. The use direction is causal earliest, with a margin reversal already at L20 (Section 3.1.2), its alignment with the output forms over the same layers (Section 3.5), and the norm of the writing direction and the feature span grow into the same band (Sections 3.4 and 3.5).

The measurements of Table 1 are correlated, because they are computed from the same activations, so their agreement at L21 marks one band of depth rather than several separate confirmations. Some onsets are weak, for example the attribution peak at 1.10 times its null, and the causal onset moves with the steering strength. The table supports the claim that commitment forms within a narrow band centred on L21, not that any single measurement is decisive on its own.

S.C Spinning of directions

The three directions are not fixed across depth. Each is recomputed at every layer, and each rotates as depth increases. This appendix measures how fast each direction rotates and shows that the three rotate at different rates.

We measure the self-alignment of a direction across a depth displacement,

$$\rho(\Delta k) = \left\langle \left| \cos(v^{(\ell)}, v^{(\ell+\Delta k)}) \right| \right\rangle_{\ell}, \tag{4}$$

the average over layers of the absolute cosine between the direction at layer ℓ and the same direction Δk layers later. It starts at 1 for $\Delta k = 0$ and decays as the direction reorganises, so a faster decay means faster rotation. One could summarise each curve by the displacement at which ρ falls to one half, but this single number is sensitive to interpolation at the threshold, so we report the full curve $\rho(\Delta k)$ instead.

On the α/β task the three decay curves separate cleanly (Figure 1). The reading direction decays fastest, from $\rho = 0.60$ at one layer to 0.24 at four. The use direction decays slowest, from 0.81 to 0.47. The writing direction lies between, from 0.74 to 0.39. The reading direction therefore rotates fastest and the use direction slowest.

This ordering holds on every concept tested (Figure 2). Across the three forms of the α/β decay corpus and the particle pairs, the self-alignment after four layers is lowest for the reading direction (0.27 to 0.34), highest for the use direction (0.41 to 0.60), and intermediate for the writing direction. The reading direction is the most reorganised and the use direction the most stable, on all of them.

The rotation is real and is not an artefact of the labelling. The reading and writing decays stay above a shuffled-label null, in which the class labels are permuted before the directions are recomputed, so the rotation is not produced by the labels. The use direction is compared instead with a structureless random-walk baseline and stays above it as well.

All three directions rotate, so rotation is a shared background of the task subspace rather than a property of one direction (Sections 3.4 and 3.5), and they separate by the rate of rotation. The reading direction, which is causally inert (Section 3.1), reorganises fastest, while the use direction, which carries the answer, is the most stable across depth.

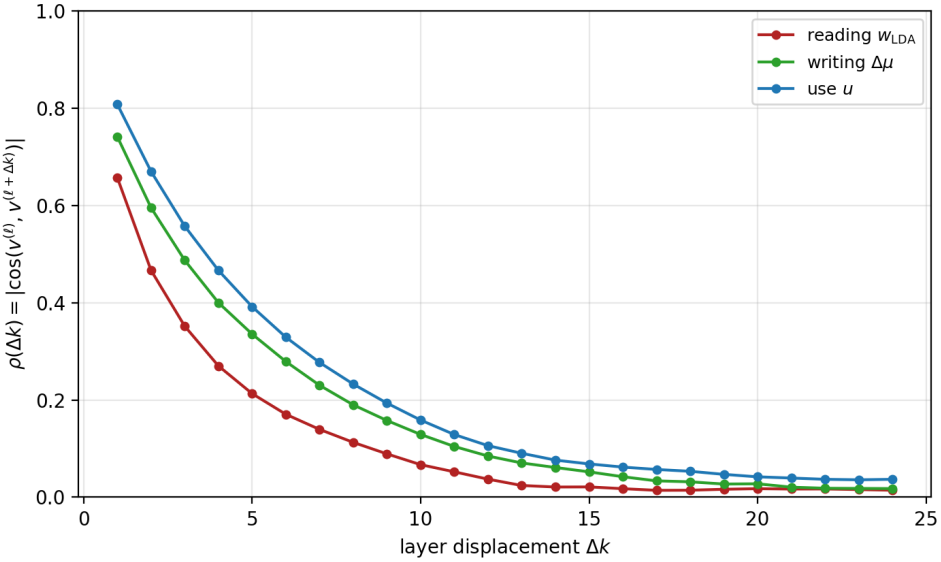


Figure 1: Rotation decay $\rho(\Delta k)$ of the three directions on the α/β task.

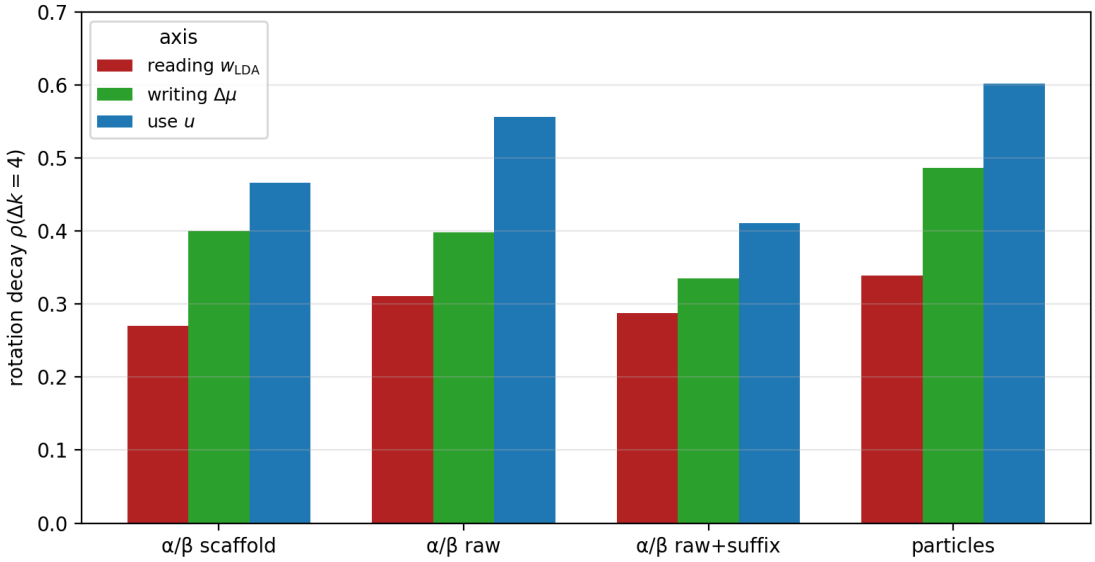


Figure 2: Self-alignment after four layers, $\rho(\Delta k{=}4)$, by concept and direction. The reading direction is lowest and the use direction highest on every concept.

S.D Geometry of clusters

This appendix describes the procedure behind the interpretation of clusters through physical features (Section 3.6), the construction of the structured cue features, the formation of the clusters, and the criteria for accepting them.

S.D.1 Prompt cue features

Each prompt of the α/β task is annotated with a set of structured physical cue features that reflect the properties of the described decay, the charge of the emitted particle (for example -1 for β^-), the change in the mass number of the daughter nucleus (for example -4 for α), the change in the charge number (for example $+1$ for β^- and -2 for α), the composition of the emitted particle (for example two protons and two neutrons for α), the presence of an antineutrino, the type of quark transition, and so on. These cue features are grouped into physical families, charge, mass, composite, lepton, and quark. The cue features serve two purposes. They provide a structured annotation for measuring cluster enrichment (Equation 20), and they separate enrichment by a physical feature from enrichment by a surface wording. Because the cue features are physically defined and do not depend on the specific wording of the prompt, the enrichment of a cluster by a cue feature (Table 7) indicates a physical feature rather than a lexical one.

A note on the confound between class and form. The surface families of wording in the α/β corpus are largely single-class (Section 2.3), so an individual cue feature correlates with the class. The enrichment of a cluster by a cue feature, for example charge -1, is therefore statistically related to the class, β. Measuring the cluster activations separately by class (Equation 21) shows, however, that the cluster features activate comparably on both classes (Section 3.6), which separates the detection of a physical feature from the detection of a class. Although the cue feature correlates with the class in the corpus, the cluster activation on it is not confined to the class label.

S.D.2 Selecting the number of clusters

The clusters used in Section 3.6 are the output of a clustering method with two choices, the algorithm and the number of clusters K. We fix both by a single requirement, that the clusters be stable under paraphrasing of the input, so that a cluster reflects a concept and not the surface wording of the prompts. We test this stability in four steps.

1. **Two paraphrase halves.** Each semantic group in the corpus appears in several wordings. We split the corpus into two halves, one holding the odd wordings and one the even, rebuild the co-importance graph on each half, and re-cluster it. Let $\mathcal{C}$ be the clustering on the full corpus and $\mathcal{C}_A$, $\mathcal{C}_B$ the clusterings rebuilt on the two halves.

2. **Recovery.** For a full cluster $G \in \mathcal{C}$ and a half clustering $\mathcal{C}'$, the recovery is the largest Jaccard overlap with any of its clusters,

$$J(G, G') = \frac{|G \cap G'|}{|G \cup G'|}, \qquad r(G) = \max_{G' \in \mathcal{C}'} J(G, G'),$$

   and the mean recovery is $\bar{r} = \frac{1}{|\mathcal{C}|} \sum_{G \in \mathcal{C}} r(G)$. A high $\bar{r}$ means the clusters survive when the wording changes.

3. **The data-reduction baseline.** A division into halves reduces the recovery on its own, since each half contains fewer prompts, independently of the wording. To isolate this reduction we also partition the corpus at random, ignoring the paraphrase structure, with the same half

sizes. The agreement between two partitions is measured by the adjusted Rand index. For the contingency counts n_{ij} of the features shared by cluster i of one partition and cluster j of the other, with row sums $a_i = \sum_j n_{ij}$, column sums $b_j = \sum_i n_{ij}$, and n features in total,

$$\mathrm{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}},$$

which equals 1 for identical partitions and is near 0 for independent ones. Over $S = 40$ random partitions we record the agreement between the two halves, $\mathrm{ARI}(\mathcal{C}_A^{(s)}, \mathcal{C}_B^{(s)})$ for $s = 1, \ldots, S$, and their distribution defines the data-reduction baseline. A paraphrase agreement at this baseline indicates that paraphrasing introduces no reduction beyond that of the smaller sample, so the clustering is invariant to wording.

4. **The stability percentile.** We locate the paraphrase agreement $\mathrm{ARI}(\mathcal{C}_A, \mathcal{C}_B)$ within the baseline distribution as a percentile, the fraction of the random partitions that it exceeds,

$$\pi = \frac{100}{S} \sum_{s=1}^{S} \mathbf{1}\left[\mathrm{ARI}(\mathcal{C}_A, \mathcal{C}_B) > \mathrm{ARI}(\mathcal{C}_A^{(s)}, \mathcal{C}_B^{(s)})\right].$$

A percentile near 50 places the paraphrase agreement at the baseline, as stable as a random partition, so the clustering is invariant to wording. A percentile near 0 places it below the baseline, so paraphrasing degrades the clusters.

We first compare the two clustering algorithms. Under the Louvain algorithm the paraphrase agreement is at the 2nd percentile of the random baseline, far below it, so its clusters change with the wording. Under average-linkage agglomerative clustering the agreement is at the 47th percentile, level with the baseline, so its clusters are stable. We therefore use agglomerative clustering.

We then apply the same test to the number of clusters, reading the percentile across a sweep of K (Table 2). At coarse resolutions, $K = 10$ and 11, the percentile is high, 87 and 90, so the clusters are stable but few. At $K = 12$ the percentile is 47, at the baseline, the finest resolution that remains as stable as a random partition. From $K = 13$ upward the percentile falls below the baseline, to 37 at $K = 13$ and to 10 at $K = 14$, so the clusters become sensitive to wording. We therefore cut the hierarchy at $K = 12$, the finest number of clusters that remains stable under paraphrasing. We could instead select K by cluster separation. A standard measure is the silhouette, the mean over features of $(b_i - a_i)/\max(a_i, b_i)$, with a_i and b_i the mean cosine distances from a feature to its own cluster and to the nearest other cluster. It rises with K throughout (Table 2) as finer clusters separate more easily, so it does not bound K, and we keep the selection by stability.

K	paraphrase percentile	silhouette
10	87	0.25
11	90	0.39
12	47	0.48
13	37	0.54
14	10	0.60

Table 2: Stability under paraphrasing across the number of clusters K, for agglomerative clustering on the co-importance graph. The percentile is the position of the paraphrase agreement in the random-partition baseline.

S.D.3 Cluster acceptance criteria

Each cluster is tested against six criteria and accepted as a candidate when it meets all of them. We write d_i for the direction along which feature i writes and $e(f, p)$ for the effect of ablating feature f on prompt p, the change it produces in the margin of the correct answer over the incorrect one.

1. **Directional consistency.** The features write with a consistent sign rather than cancelling, measured by the mean pairwise signed cosine of their directions,

$$\bar{c}_{\text{sgn}} = \text{mean}_{i<j} \cos(d_i, d_j) \geq 0.50.$$

2. **Coherence.** The features share a common axis, whether they add or oppose along it, measured by the mean pairwise absolute cosine,

$$\bar{c}_{\text{abs}} = \text{mean}_{i<j} |\cos(d_i, d_j)| \geq 0.85.$$

   A high absolute cosine with a low signed cosine marks a cluster whose features share an axis but push against each other.

3. **Semantic specificity.** The cluster is enriched for at least two physical cue features, each with a lift (Equation 20) above 1.5 for at least three values of k.

4. **Cross-form reuse.** The cluster is active across several wordings of the same concept rather than one. A composite reuse score of at least 0.40 combines the diversity of the cluster's effect across the complexity levels, the prompt groups, and the cue types with its balance between keyword-free and keyword-present prompts.

5. **Causal impact.** The joint ablation of the cluster changes the answer or the geometry. The cluster passes if the interaction ratio

$$\text{IR}(C, p) = \frac{e_{\text{joint}}(C, p)}{\sum_{f \in C} e(f, p)},$$

   the effect of ablating all its features together over the sum of their individual effects, is at most 0.50, or if the fraction of prompts whose answer flips under joint ablation exceeds 0.05, or if a test against additivity gives $p < 0.001$.

6. **Level coverage.** The cluster is active for prompts at at least three of the four complexity levels.

Of the twelve clusters of the α/β task (HAC $K = 12$), six are accepted candidates that meet all six criteria including the causal one, three are partial candidates, the multilayer hub whose coherence falls below threshold together with two clusters that meet the five local criteria but fail the causal criterion, their joint ablation flipping fewer than five percent of prompts, and three are descriptive only, the single feature clusters. This ensures that the clusters on which the writing account rests (Section 3.6) are coherent, causal, and functional groups rather than artefacts of the partitioning.

To link the latent organisation of features with the trichotomy of directions, we measure how the ablation of a cluster changes the geometry of the writing and use directions. Ablation is performed by direct subtraction from the true MLP output, after which the change of the $\Delta\mu$ and u directions across layers is recorded against a null distribution built from random groups of the same size and from the same layers.

S.D.4 Cluster ablation

The change of a direction under ablation is measured by the norm-free angle

$$\theta^{(\ell)} = \arccos\left(\langle \hat{v}^{(\ell)}_{\text{before}}, \hat{v}^{(\ell)}_{\text{after}} \rangle\right),$$

the angle between the unit direction before and after ablation.

Ablation of clusters rotates the writing direction $\Delta\mu$. The change in norm is small and always corresponds to compression rather than growth. On the layers where the ablation has an effect, that effect is almost entirely a rotation, about 99 percent of it, and not a change in norm.

The strongest rotations are produced by the early clusters. The charge cluster, at L10 with the rotation measured across L10 to L13, rotates the writing direction by about 14.6 degrees and the lepton number cluster by about 12.7 degrees. Intermediate rotations come from the late output cluster, about 9.2 degrees, and the cluster at L24, about 8.2 degrees. Single clusters at the intermediate layers produce small rotations of about 1 to 3 degrees. This pattern, in which the early clusters rotate the writing direction most, is consistent with the writing being formed early (Section 3.2) and used causally later (Section 3.1).

At most layers the ablation of a single feature changes the writing direction by about 1 degree, in the range 0.8 to 1.7 degrees, so the writing direction is largely distributed and is not localised to any single feature. The null for single features is not reliably estimated because the pool of single feature groups is small.

S.D.5 Writing direction rotation

In the α/β decay corpus the writing and use directions are close to orthogonal, with $\cos(\Delta\mu, u)$ about 0.008, unlike on the scaffold form (Section 3.4) where they are partly aligned. Cluster ablation rotates the writing direction by up to about 15 degrees (Section S.D.4), but this rotation is not directed toward the use direction, and therefore not toward the output, since the use direction converges to the unembedding (Section 3.1). The change it produces in the alignment of the two directions, $\mathrm{d}\cos(\Delta\mu, u)$, lies between about -0.009 and $+0.019$ across the clusters and does not exceed the null covariance, whose 95th percentile is about 0.037 to 0.039. The strongest change toward use, from the early charge cluster at about $+0.018$, is still below this threshold, and the remaining clusters give changes of order ± 0.001 to 0.006 with no stable sign. The near orthogonality is therefore preserved, the rotation takes place in components other than the use direction, and the separation of writing and use is not held up by any single cluster whose removal would destroy it.

This is consistent with the convergence of the writing direction to the output (Section 3.1) rather than in tension with it. That convergence is a depth-distributed process not attributed to individual clusters, matching the distributed picture of Section S.D.4, in which the writing direction is not carried by any single feature.

S.E Whitening turns the optimal probe from the use direction

Section 3.1.3 established that the optimal linear decoder w_{LDA} is almost orthogonal to the use direction u. Here we show that this orthogonality follows from the definition of the probe. The probe is $w \propto \Sigma^{-1}\Delta\mu$ of Equation 7, where $\Delta\mu$ is the difference of class means and Σ is the within-class covariance with the diagonal regularisation of Section 2.5.2. All quantities are computed on the train split with the same regularisation to which the probe is fitted. For a unit direction $\widehat{u}$ we write $\sigma_u^2 = \widehat{u}^\top \Sigma \widehat{u}$ for the variance of the representation along $\widehat{u}$.

Let $\widehat{u}$ first be an eigenvector of Σ with eigenvalue σ_u^2. Using $\langle \Sigma^{-1}\Delta\mu, \widehat{u}\rangle = \langle \Delta\mu, \Sigma^{-1}\widehat{u}\rangle = \sigma_u^{-2}\langle \Delta\mu, \widehat{u}\rangle$ and dividing by $\|\Sigma^{-1}\Delta\mu\| = \|\Delta\mu\|/\lambda_{\mathrm{eff}}$,

$$\cos(w, \widehat{u}) = \cos(\Delta\mu, \widehat{u})\, \frac{\lambda_{\mathrm{eff}}}{\sigma_u^2}, \qquad \lambda_{\mathrm{eff}} = \frac{\|\Delta\mu\|}{\|\Sigma^{-1}\Delta\mu\|}. \tag{5}$$

The alignment of the probe with an axis equals the alignment of the raw difference of means divided by the variance along that axis. For a general unit $\widehat{u}$ that is not an eigenvector, set $r = \Sigma\widehat{u} - \sigma_u^2\widehat{u}$. Then $\Sigma^{-1}\widehat{u} = (\widehat{u} - \Sigma^{-1}r)/\sigma_u^2$, and substitution into the second step gives the exact equality

$$\cos(w, \widehat{u}) = \cos(\Delta\mu, \widehat{u})\, \frac{\lambda_{\mathrm{eff}}}{\sigma_u^2} - \frac{\langle \widehat{w}, r\rangle}{\sigma_u^2}. \tag{6}$$

The first term is a prediction and the second a computable residual. The scale λ_{eff} is small, since in the eigenbasis of Σ one has $\|\Sigma^{-1}\Delta\mu\|^2 = \sum_i d_i^2/\lambda_i^2$ with d_i the coordinates of $\Delta\mu$, so λ_{eff} is a weighted power mean of order minus two of the eigenvalues and is dominated by the smallest of them. After whitening the probe concentrates in the low-variance subspace, and the ratio $\lambda_{\mathrm{eff}}/\sigma_u^2$ is small for any axis with appreciable variance.

The use direction is such an axis. Its variance sits at the 98.4th and 98.7th percentile of the spectrum of Σ on the two concepts, the factor $\lambda_{\mathrm{eff}}/\sigma_u^2$ in the decision zone is 0.059 on the decay task and 0.031 on number agreement, and the prediction of Equation 5 agrees with the measured alignment to within the residual of Equation 6. The per-layer prediction error is 0.039 and 0.044, and the measured residual $\langle \widehat{w}, r\rangle/\sigma_u^2$ is 0.038 and 0.048, so the prediction error is the residual term of the exact equality (Table 3).

concept	$\cos(\Delta\mu, u)$	percentile of σ_u^2	$\lambda_{\mathrm{eff}}/\sigma_u^2$	prediction error	residual
decay α/β	0.28	98.4	0.059	0.039	0.038
number agreement	0.18	98.7	0.031	0.044	0.048

Table 3: The attenuation identity in the decision zone, medians on the train split. The alignment $\cos(\Delta\mu, u)$ before whitening, the percentile of the variance σ_u^2 along u in the spectrum of Σ, the attenuation factor of Equation 5, the per-layer error of that prediction, and the exact residual of Equation 6. The prediction error equals the residual, so the identity holds to the residual term.

The variance along u is large for the following reason. The margin is a nonlinear function of the layer state, $\mathrm{margin}(p) = M(h(p))$. A first order expansion about the class mean gives $\mathrm{margin}(p) \approx M(\mu_c) + \langle \nabla M(\mu_c), h(p) - \mu_c\rangle$, and if the gradient does not vary from prompt to prompt, $g(p) \approx \bar{g}$, the within-class fluctuation of the margin is the projection of the state onto $\bar{g}$. The variance of that projection is a quadratic form of the covariance,

$$\mathrm{Var}_c(\mathrm{margin}) = \bar{g}^\top \Sigma\, \bar{g} = \|\bar{g}\|^2 \sigma_u^2, \qquad \widehat{u} = \bar{g}/\|\bar{g}\|. \tag{7}$$

The within-class fluctuations of the confidence of the model lie along the direction by which it reads the answer, and that direction therefore carries a large within-class variance. Both approximations

are checked by the ratio $\mathrm{Var}_c(\mathrm{margin})/(\|\bar{g}\|^2\sigma_u^2)$, which is one where they hold. At the readout layer it is 1.06 and 1.16 and Equation 7 is exact, while in the decision zone it rises to 20.1 and 3.9, since the gradient varies from prompt to prompt and adds variance to the margin beyond the one accounted for, so in the zone Equation 7 gives a lower component of the margin variance and the high variance of u rests on the measured percentile.

The result is as follows. The exact equality of Equations 5 and 6 turns the optimal probe from any high-variance axis by a computable factor, the measurement shows that the use direction is such an axis, and Equation 7 explains why. The use of a direction is accompanied by variance along it, which hides it from the optimal linear reader.

A remark on the regularisation. The weight $s = 0.1$ is fixed a priori and shared by all analyses. As it is raised the attenuation weakens toward the ratio of the diagonal scales, so the effect is not created by the regularisation, and the value reported at $s = 0.1$ is a conservative one that an unregularised probe would only strengthen.

S.F Reproduction of circuit tracing

S.F.1 Staging and pipeline

The first phase of the project reproduced the attribution-graph methodology of [3] and the analytical style of [4] on an open model, Qwen3-4B (base) with per-layer transcoders [5] in place of the cross-layer transcoders of the original. The pipeline ran in five stages. We fixed the answer position and ran the baseline, extracted the active transcoder features at that position, built an attribution graph whose edges are the gradient-times-activation attributions between features, validated its nodes and edges by feature ablation and activation patching, and assembled family chains for comparison across the surface framings. The intervention protocol and the fixing of the answer position are the canonical setup used throughout the main text.

S.F.2 What the reproduction revealed

The methodology transferred to the open model. The chains are compact, 13 to 26 features and 19 to 60 edges, and their nodes carry readable object-level content. Three observations of this phase then shaped the programme of the main part.

Distribution with low necessity. The necessity of individual paths is systematically low, 10 to 28% across corpora, while the total causality of the chain is high, so the causal structure is a property of the ensemble rather than of the listed units. This became the distributed, non-localised picture of Section 3.6 and Appendix S.D, and it fixed the cluster, rather than the feature, as the minimal causal unit.

The attribution label is not the causal direction. A sign-complete ablation, over all features rather than the positive ones alone, revealed a systematic divergence. The β-prompt sign-flip rate at layer 24 is 25.4% against 0.4% on α, which the attribution labels do not predict. The labels proved an unreliable indicator of causal role, and we set them aside in the main part (Section 3.6) in favour of the directly measured geometric objects, the reading, writing, and use directions.

A chain can be present while its output is suppressed. The circuits converge across levels (interchange similarity at least 0.74, including levels of zero accuracy), and some carrier features hold a near-constant activation yet a non-zero causal effect ($\Delta\text{act} \approx 0$, $\Delta\text{eff} = +1.16$), so the presence and readability of a structure do not predict its contribution to the answer. This is an early form of the central dissociation between what is decoded and what is used (Sections 3.1.3 and 3.5).

Taken together, the reproduction succeeded as a method and proved insufficient as an explanation. The attribution graphs answer through which nodes the computation flows, but not the geometry in which the decision is read, written, and used. The main part answers this second question.

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